

DAX (Data Analysis Expression)

⇒ It is the formula in power BI. There are three types of DAX.

- (I) Table DAX
- (II) Column DAX
- (III) Measure DAX

(I) TABLE DAX

⇒ Example I am having a table which contains P.ID, P.Name, Sales, Cost price.

Now, I want a profit also but on the new table not on the existing table so I will use Table DAX.

It gives the entire information new table with the entire information without affecting the original table.

- I want a ~~country~~ country = Germany in new table → Table view
→ Table tools → New table → Germany Country = FILTER
(financials, financials [Country] = "Germany") → Enter
- Now I want Country = Germany and Product ~~value~~ sales.

Table view $\rightarrow$ Table tools $\rightarrow$ Country Germany & Product velo
= Calculate table (financials, financials [Country] = 'Germany',
financials [Products] = "velo")

- Now I need country = Germany OR Canada, Product = velo and financials sales must be greater than 50,000
- Table view $\rightarrow$ Table tools $\rightarrow$ New table $\rightarrow$ Country Germany or Canada & product velo and sales greater than 50,000
= Calculate table (financials, financials [Country] = 'Germany' || financials [Country] = "Canada", financials [Products] = "velo" & financials [Sales] > 50,000).
- Now I want year wise country wise financial total sales.

@ In ~~the~~ PBI DAX group by is known as summarize.

group-By = summarize (financials, financials [Year],
financial [Country], "Total Sales", sum (financials [Sales]))

Now I need data of year 2013 only from the group by table.

Year 2013 = filter (summarize (financials, financials [Year],
financials [Country], "Total Sales", sum (financials [Sales])),
financials [Year] = 2013)

Calculate table

$\rightarrow$ Filter & Filter function is use for the simple analysis only
when we to handle complex analysis that time we use
calculate table function.

Top N Dax

⇒ It provide N number of top value.

Example ⇒ From the last group by table now I need a top sale.

Top sales = Top N (1, group-by, group-by [Total sales])

Calendar & Calendar auto

⇒ In the Calendar auto the system automatically bring the date from the key its own mind.

where in Calendar dax we have to put the dates manually.

Date = Calendarauto()

Calendar = Calendar (min(financials [Date], max(financials [Date])))

To append : Union function is used here.

Append = union (Accounts - Data 1, operations - Data 2).

Column DAX

⇒ Same table with new column.

⇒ It provides the additional information by creating a new column.

Table view → Table tools → New Column

Example ⇒ I am having order-date and ship-date in my Indian Data, Now I want to calculate the difference between to dates.

Age = ABS (DATEDIF (Indian - Data [Ship - Date], Indian - data [order - date], Day))

* Here we using ABS function to make all value positive.

Bin / Container / List data groups:

→ Select the age column → Column tools → Data groups →
New data groups → Group type (list)

How to extract the data, day, month, month no, Quarter, Year?

→ Table tools → New Column → Month name = Indian date
[order date].[Month]

Month Name = FORMAT (Indian date [order Date], "mmmm")

@ FORMAT IS USED OR WORK AS TEXT

My notes to Distinguish

→ Just we used calculate there to sum the Telangana sales, it doesn't mean that calculate works only for sum x. it can work for sum x similarly for other functions also.

Its main work is to focus on measure and then applying the filter.

There we did focus on "Total sales" and then kept filters state = "Telangana".

So, from the total sales it provided the sum of Telangana only.

Power BI default its working as sum x but don't think that its on sum x.

→ Calculate DAX works only for measures not for column level.
→ We can some times use CALCULATE with other x functions instead of Average x, sum x etc, but it depends on what you are analyzing.

Difference between SUM x & Calculated DAX?

→ Use sum x when you need row level computation.
→ Use Calculate when you need filtering.